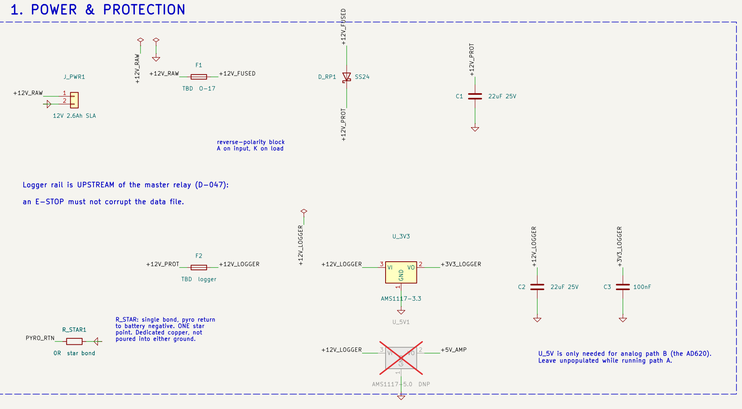
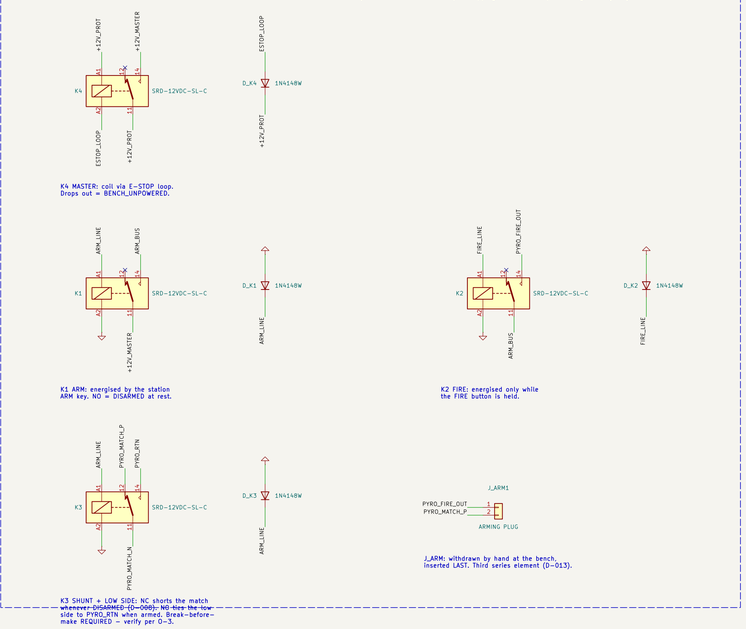
1. POWER & PROTECTION



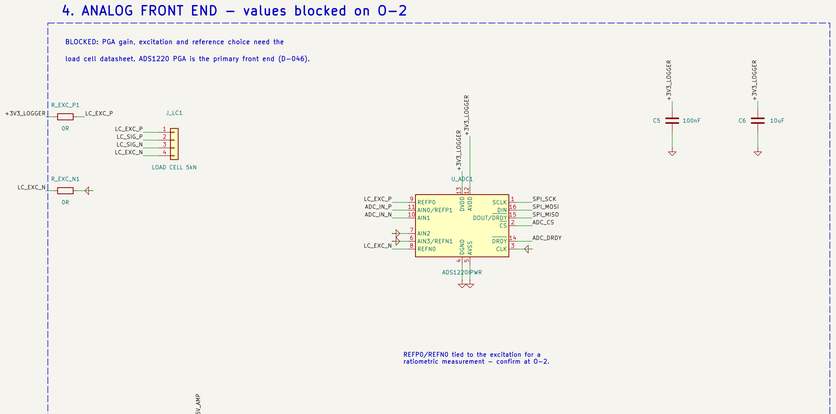
2. SAFETY CHAIN — hardware only, no MCU



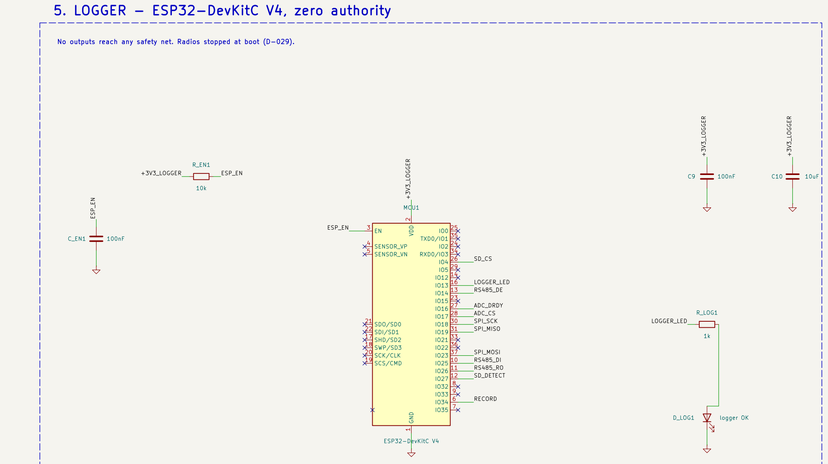
3. PYRO INTERFACE — values blocked on O-1



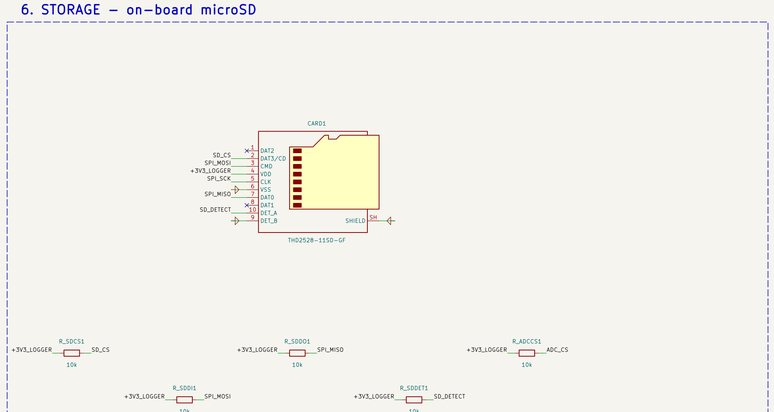
4. ANALOG FRONT END — values blocked on O-2



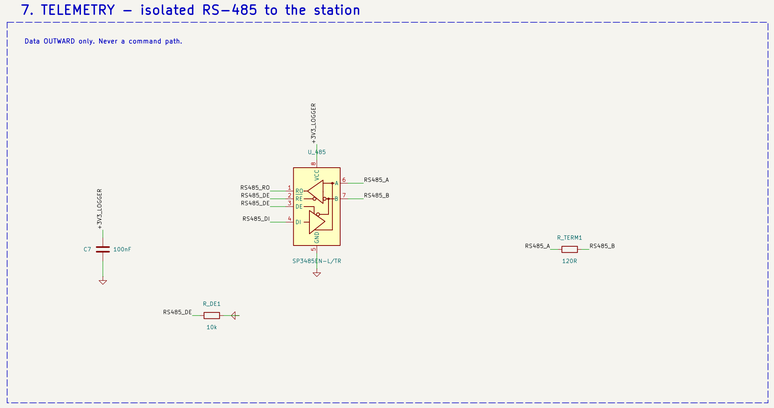
5. LOGGER — ESP32-DevKitC V4, zero authority



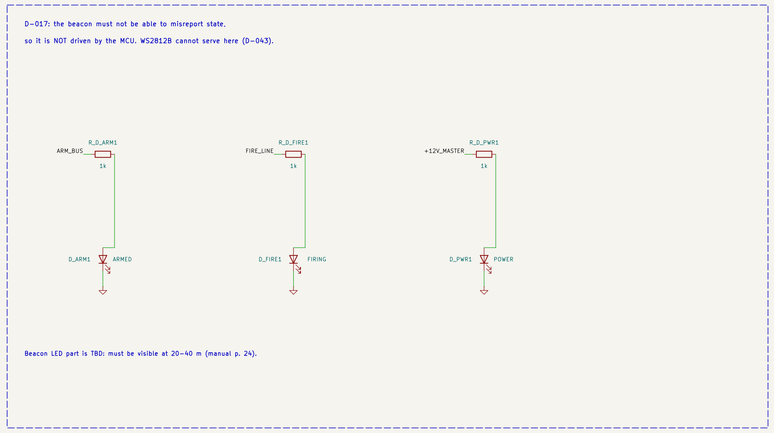
6. STORAGE — on-board microSD



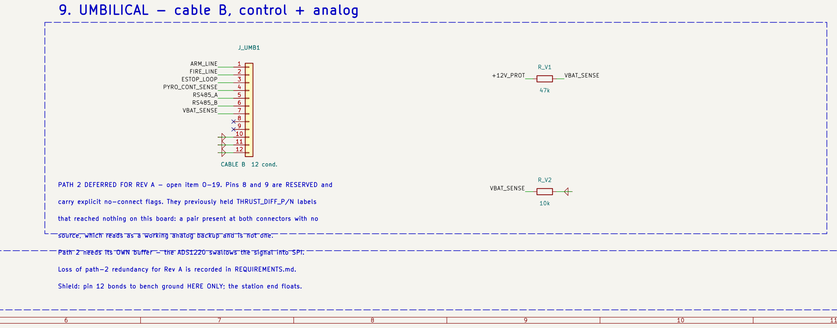
7. TELEMETRY — isolated RS-485 to the station



8. INDICATORS — beacon driven from the REAL arm/fire lines



9. UMBILICAL — cable B, control + analog



TBD values await O-1 (e-match no-fire current) and O-2 (load cell datasheet).

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_bench.py
Igniter safe state is SHORTED via K3 normally-closed. Values marked TBD await O-1 / O-2.
PRELIMINARY — NOT APPROVED. NO MCU on the arm, fire or e-stop path.

Ignitia Rocket Lab	
Sheet: /	
File: talavera-bench.kicad_sch	
Title: TALAVERA v1 — Bench Board	
Size: A1	Date: 2026-11-10
KiCad 10.0.0	Rev: A.1
	Id: 1/1